

HalftimeHash: a repaired collision analysis and bounds for the shipped code

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With machine-assisted analysis

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Abstract

HalftimeHash’s published composition proof incorrectly treats noncollision of a vector as noncollision in every coordinate. Its ring argument also confuses a power of two with its exponent and omits a division. We repair the intended construction using matrix fibres and a projection-polynomial inequality. For a distance- k encoder the resulting bound is $2^{-32k}(h+2)^{k-1}(h+1+2^t)$, where t is the maximum determinant valuation. This recovers the intended HalftimeHash24 guarantee of more than 83 bits at height 16. The code at commit `caf7924` is a different matter: its distance-3 encoder has distance 1, and a fixed equal-length pair collides with probability at least 2^{-32} . The public Style wrappers use the valid distance-2 encoder. On their explicit stack-safe domains, and for their flat-address, unsigned-arithmetic interpretation with independent uniform key words, their overlapping key layout admits the optimal length-normalised coefficient $2-2^{-64}$, giving a conservative 63-bit certificate. These are neither portable-C++ guarantees for the defective table type nor claims about PRNG-expanded keys.

Method and evidence. This note consolidates several independent automated passes and execution checks: the audit, a proof note, an adversarial referee report, an independent execution report, and two further condensed reviews. All mathematical statements retained as proved below were re-derived, and numerical and implementation statements were re-derived or executed rather than taken on trust. Exact matrix arithmetic and focused real-header checks were rerun for this draft; the large historical sampling runs are reported as supplied evidence, not claimed as rerun. The proofs below are written out afresh. Machine assistance is not formal verification: missing primary review packages and logs are marked explicitly, and Section 6 distinguishes execution from Lean checking. Source conflicts are resolved in Appendix C.[2, 9, 10, 4, 11, 5]

1 The construction in the paper

We follow Apple’s notation for arXiv:2104.08865v2, with one deliberate change: t denotes a determinant’s maximum *valuation*, and $P = 2^t$ the corresponding power of two. Let

$$Q = 2^{32}, \quad R = 2^{64} = Q^2, \quad \varepsilon = Q^{-1}, \quad \delta = R^{-1} = \varepsilon^2.$$

All probabilities are over independent uniform key words; the two distinct messages are fixed before the key is chosen. Arithmetic on hash outputs is in $\mathbb{Z}_R$. A family is η -AU if collision probability is at most η , and η -AΔU if, for every distinct pair and every prescribed $c \in \mathbb{Z}_R$, the probability of output difference c is at most η . Vector AΔU prescribes every coordinate. These are collision statements, not claims that output is uniform.[1, 9]

A block contains b 64-bit lanes. An EHC group consists of d symbols, each of w blocks. The encoder J produces e symbols and is intended to have minimum *symbol* distance k . Independently keyed NH hashes compress each symbol to a block, and a $k \times e$ integer matrix T combines the results, lane by lane modulo R :

$$\text{EHC}(x) = T(H_0(J(x)_0), \dots, H_{e-1}(J(x)_{e-1}))^\top.$$

NH on a 64-bit pair of 32-bit halves uses

$$N_{s,t}(x, y) = ((x + s) \bmod Q)((y + t) \bmod Q) \pmod{R}; \quad \text{NH} = \sum_i N_{s_i, t_i}(x_i, y_i). \quad (1)$$

It is ε -A Δ U on equal-length distinct strings. Tree nodes use the Badger modification: pass through the first child and NH-hash the remaining children. This is ε -AU; if only the passed-through child differs, collision is impossible.[1, 3, 9]

For completeness, the one-pair NH argument includes the wrap. Condition on one key half so that the factors associated with a differing input half are distinct $c, c' \in [0, Q)$. The remaining equation is $cz - c'((z + d) \bmod Q) = v \pmod{Q^2}$, with z uniform in $[0, Q)$. On each side of the wrap its nonzero slope gives at most one solution. Two solutions separated by the wrap would give $(c - c')(z_2 - z_1) + Qc' \equiv 0 \pmod{Q^2}$; this integer lies strictly between 0 and Q^2 . Thus there is at most one solution in total. Condition on all other pairs to obtain the sum version.[9, 10]

Each of the k EHC coordinates feeds a forest of complete f -ary trees. For the intended construction, write the number N of full groups in ordinary base f and use the digit at height j as the number of roots of height j . Put $h = \lfloor \log_f N \rfloor$ for $N \geq 1$, and $r_f(N)$ for the digit sum, so $r_f(N) \leq (f - 1)(h + 1)$. Fresh final NH hashes the ordered roots in each coordinate and sums its lanes. A separate Toeplitz NH pool hashes the raw tail to k outputs. The final result is the sum of forest and tail contributions.[11, 5]

Keys may be reused across EHC groups and across SIMD lanes. Tree keys may be reused among nodes at the same height. They must be independent across stages, heights and output coordinates; forest-finalizer keys are distinct for each root, coordinate and lane. Toeplitz tail outputs share shifted windows of a separate pool. No independence among EHC output coordinates is assumed. Equal-length inputs have the same forest shape, tail layout and key offsets. An explicit length tag can extend the intended core to unequal lengths; the shipped wrappers require their own analysis.[2, 5, 9]

2 What is wrong in the published proof

Lemma 3: a vector can differ in only one coordinate. The proof multiplies the k coordinate-tree bounds after assuming EHC has not collided. But $(0, 0)$ and $(1, 0)$ are distinct vectors whose second coordinate is identical. For a height-one NH comparison, zero versus one in the first half gives products st and $((s + 1) \bmod Q)t$. These agree exactly when $t = 0$, with probability ε . The second coordinate collides with probability one. Independent tree keys therefore give ε , not ε^2 . The same obstruction applies with more coordinates. This refutes the conditional inference; it does not by itself refute the intended construction's final theorem, because its EHC distribution has additional structure.[2, 6]

Theorem 1: valuation, division, and inequality. The paper defines p as a power of two but uses it as an exponent, including $2^{k(p-32)}$. Write $\det A = q2^{t'}$, with q odd and $t' \leq t$. Multiplication by the adjugate yields a necessary condition, not generally an equivalent one. After removing q , the equation $2^{t'}Z = \beta \pmod{R}$ has no solution unless β is divisible by $2^{t'}$; when it is, the correct condition is

$$Z \equiv \beta/2^{t'} \pmod{2^{64-t'}}.$$

The paper omits that division. Counting lifts and applying A Δ U gives an upper bound, not the printed equality. A repaired adjugate argument gives $P^k \varepsilon^k$; the next section improves it to $P \varepsilon^k$. Smaller errata appear in Appendix A.[11, 2, 6]

3 A repaired theorem for the intended construction

Lemma 3.1 (Matrix fibres). *Let $A \in \mathbb{Z}^{r \times r}$ be nonsingular. Write its integer Smith invariants as $d_1, \dots, d_r$. Over $(\mathbb{Z}_{2^{64}})^r$, every nonempty fibre has size*

$$|\ker A| = \prod_i 2^{\min(64, \nu_2(d_i))}.$$

In particular, if $\nu_2(\det A) < 64$, this is $2^{\nu_2(\det A)}$.

Proof. Integer Smith normal form gives unimodular U, V with $UAV = \text{diag}(d_i)$. Their determinants are ± 1 , so both changes of variables remain invertible modulo 2^{64} . The scalar equation $d_i z = 0$ has $\gcd(d_i, 2^{64})$ solutions. Multiply these counts; a soluble fibre is a translate of the kernel. The restriction in the last sentence matters: the unqualified determinant formula is false for arbitrarily large valuations.[9, 11, 5] $\square$

Fix k differing encoded symbols F , possible by the distance assumption. For a row subset $S \subseteq [k]$, define

$$c_{S,F} = \min_{\substack{J \subseteq F, |J|=|S| \\ \det T[S,J] \neq 0}} 2^{\nu_2(\det T[S,J])}, \quad c_{\emptyset,F} = 1. \quad (2)$$

Full rank ensures the minimum exists. Conditioning on all symbol keys except those indexed by a minimizing J makes the selected symbol differences independent, with every joint atom at most $\varepsilon^{|S|}$. Lemma 3.1, or its upper bound when a valuation is large, therefore gives

$$\Pr[\text{EHC}(x)_S - \text{EHC}(y)_S = z_S] \leq c_{S,F} \varepsilon^{|S|}. \quad (3)$$

For block outputs, choose a physical lane in which the input group differs. The encoder must have distance k on that lane. Equality of whole blocks implies equality on this lane; broadcast keys cause no difficulty.[2, 9]

Theorem 3.2 (Corrected EHC theorem). *Suppose every k -column submatrix T_F is nonsingular, and $t = \max_F \nu_2(\det T_F) < 64$. With a distance- k encoder and independent ε - $A\Delta U$ symbol hashes, EHC is $2^t \varepsilon^k$ - $A\Delta U$. In particular its bound for NH is 2^{t-32k} .*

Proof. Use all rows in (3). This conditions on a chosen differing group only; key reuse across other groups does not enter the claim.[11, 5] $\square$

Lemma 3.3 (Projection-polynomial inequality). *For any nonsingular integer $A = T_F$ with $\tau = \nu_2(\det A)$ and any $u, v \geq 0$,*

$$\sum_{S \subseteq [k]} c_{S,F} u^{k-|S|} v^{|S|} \leq (u+v)^{k-1} (u+2^\tau v). \quad (4)$$

Proof. Integer unimodular column operations triangularise A : $B = AV$ is upper triangular with nonzero diagonal, $V \in \text{GL}_k(\mathbb{Z})$. This follows recursively from the Euclidean algorithm on the last row, leaving only its last entry nonzero, then treating the upper-left submatrix. Put $e_i = \nu_2(B_{ii})$. They are nonnegative and sum to τ .

For each S , the principal minor $\det B[S, S]$ is the product of the selected diagonal entries, of valuation $\sum_{i \in S} e_i$. Cauchy–Binet expresses it as

$$\det B[S, S] = \sum_{|J|=|S|} \det A[S, J] \det V[J, S].$$

If every nonzero A -minor in that sum had larger valuation, their integer linear combination would too. Hence $c_{S,F} \leq 2^{\sum_{i \in S} e_i}$, and the left side of (4) is at most $\prod_i (u + 2^{e_i} v)$. Finally, for $a, b \geq 1$,

$$(u+v)(u+abv) - (u+av)(u+bv) = uv(a-1)(b-1) \geq 0.$$

Repeatedly replacing two factors by ones with multipliers 1 and their product concentrates the entire valuation in one factor, proving the claim. This is a general lemma, not an enumeration of the four matrices.[5] $\square$

TODO: Obtain the fourth review's full PDF/TeX and verification package; the projection proof here is independently reconstructed from its condensed description, not quoted from an available full proof.

Lemma 3.4 (One-component forest). *For two different fixed sequences with the same complete-tree forest shape and maximum height h , the ordered root lists agree with probability at most $h\varepsilon$. Adding independently keyed final NH gives at most $(h+1)\varepsilon$ for any prescribed output difference. More sharply, the two bounds are $1 - (1-\varepsilon)^h$ and $a_h = 1 - (1-\varepsilon)^{h+1}$.*

Proof. Select a complete tree whose leaf sequences differ. At each of its levels, conditional on all lower-level keys and survival of a difference, select a differing child tuple. The fresh key for that height merges it with probability at most ε . Thus a difference survives to a root with probability at least $(1-\varepsilon)^h$; equality of all roots implies equality of this tree's root. Reuse at the same height is harmless because only one differing tuple is selected. If the root lists still differ, condition on their values and all but one differing pair's fresh final-NH key. The NH difference bound applies to any target, including after summing other lanes. Combining these events gives $a_h \leq (h+1)\varepsilon$. [2, 9, 5] $\square$

Theorem 3.5 (End-to-end intended bound). *Under the distance, matrix, forest and key assumptions in Section 1, the collision probability for equal-length distinct inputs with at least one full group is at most*

$$\min\{1, 2^{-32k}(h+2)^{k-1}(h+1+2^t)\}. \quad (5)$$

With no full EHC group it is at most ε^k .

Proof. First assume equal raw tails and choose a differing group. Let I_j indicate equality of its complete EHC output block in coordinate j . Conditional on all EHC keys, if the whole coordinate sequence is identical then $I_j = 1$; otherwise Lemma 3.4 bounds its collision probability by $\rho = (h+1)\varepsilon$. The coordinates' remaining tree and root-finalizer pools are independent. If $\rho \geq 1$, the clipped theorem is trivial. Otherwise,

$$\begin{aligned} \Pr[\text{collision}] &\leq \mathbb{E} \prod_{j=1}^k (\rho + (1-\rho)I_j) \\ &\leq \sum_{S \subseteq [k]} c_{S,F} \rho^{k-|S|} \varepsilon^{|S|} \\ &\leq (\rho + \varepsilon)^{k-1} (\rho + 2^t \varepsilon), \end{aligned}$$

by (3) and Lemma 3.3. No independence of the I_j was used.

If the tails differ, condition on all non-tail keys. The required tail difference is now a fixed vector. Flatten the tail into 64-bit NH pairs, and choose its last differing pair. In the last output window its key is beyond every key relevant to the earlier output differences. The one-pair argument costs ε conditional on all the other keys. Repeat backwards through the k shifted windows to get ε^k . Identical trailing pairs cancel, including padding. This proves the tail case and the no-group case. It is conditioning on a separate case, not an extra additive error term. [2, 9, 5] $\square$

The matrices and numerical specialisations are as follows. Their maximal minors were checked in exact arithmetic in the supplied reviews and rechecked for this draft by recursive determinants and rational elimination. All are nonsingular; the counts are $21 + 84 + 210 + 126 = 441$. [11, 6] For $k = 3$, direct subset

Core bytes / k	(d, e)	t	EHC only	Bound including forest and tail
16 / 2	(6, 7)	2	2^{-62}	$2^{-64}(h+2)(h+5)$
24 / 3	(7, 9)	2	2^{-94}	$2^{-96}(h+2)^2(h+5)$
32 / 4	(7, 10)	3	2^{-125}	$2^{-128}(h+2)^3(h+9)$
40 / 5	(5, 9)	3	2^{-157}	$2^{-160}(h+2)^4(h+9)$

Table 1: Intended-construction bounds, clipped at one. Here $w = 3$, and the implementation uses $f = 8$. Parameters and values: [1, 11, 5]. Matrices are reproduced in Appendix B.

enumeration also gives sums 6, 9, 4 for subsets of sizes 1, 2, 3, respectively, agreeing with $(u+v)^2(u+4v)$. [2]

Corollary 3.6. *For the intended HalftimeHash24 construction at $h \leq 16$, the collision probability is at most $6804 \cdot 2^{-96}$, giving $96 - \log_2(6804) \approx 83.27$ bits.*

Proof. At $h = 16$, $(h+2)^2(h+5) = 18^2 \cdot 21 = 6804$. For the paper's $b = 8, d = 7, w = 3, f = 8$, an EHC group occupies 168 64-bit words; inputs below the paper's exabyte scale have height at most 16. This corollary assumes a correct distance-3 encoder and adequate forest storage and key allocation. [11, 5, 2] $\square$

4 Bounds for the shipped code

All source claims in this section concern commit `caf7924ceab4721f4e0cc33442b185558ba7f1c4`, identified also by the header digest in Appendix C. The execution model has eight-bit bytes, eight-byte `size_t`, little-endian loads, intended flat addresses, and modular unsigned sums. This is the tested arithmetic function, not a claim about every ISO-C++ optimisation of the literal source. [9, 7, 5]

4.1 The encoder defect and its collision witness

Write the raw blocks as $r_0, r_1, \dots$. In `Encode3`, `DistributeRaw` captures `iter` by value at `raw_io + 1`; subsequent outer-pointer changes do not change its reads. Pointer increments advance one block, not one three-block symbol. Symbolic XOR evaluation gives exactly

$$J(x)_7 = (r_0, r_1, r_2), \quad J(x)_8 = (r_0 \oplus r_1, r_1 \oplus r_2, r_2 \oplus r_3). \quad (6)$$

The first parity loop repeats the frozen triple an even 26 times, and the remaining transforms produce the second expression. Thus changing only r_6 changes only systematic symbol 2. Systematicity excludes distance zero, so the minimum symbol distance is exactly 1. The same capture defect affects `Encode4` and `Encode5`: all added packets depend only on the first four raw blocks, so the same changed block gives distance 1.[4, 9, 5]

For $b \in \{1, 2, 4, 8\}$, compare 168 b zero bytes with the same message except that word $6b$ equals 1. This is one full width-3 group, differing only at lane zero of block 6. Write the responsible key word as `core_key`[6] = $A + QZ$. The changed symbol-NH difference is

$$\Delta = \begin{cases} Z, & A < Q - 1, \\ -(Q - 1)Z \pmod{R}, & A = Q - 1. \end{cases} \quad (7)$$

Both multipliers are odd; hence $\Delta = 0$ iff $Z = 0$. Column 2 of T_3 is $(1, 0, 1)^\top$, so this is also exactly the EHC-collision condition. Every subsequent stage receives identical values when it holds. Of the Q^2 values of this key word, exactly Q have $Z = 0$, regardless of all other keys. Consequently

$$\Pr[\text{full 24-byte core collision}] \geq 2^{-32}.$$

This is an exact count of a sufficient event. Later stages can introduce additional collisions; we do not claim an exact full-hash probability or a full-hash “iff”. Padding the same witness with identical groups extends the lower bound to every stack-safe equal length containing a full group. For $k = 4, 5$ use their own group sizes and the same changed block.[9, 4, 5]

b	Witness bytes	Unconditioned hits / 2^{34}	Forced $Z = 0$	Forced $Z \neq 0$
1	168	4	$2^{20}/2^{20}$	$0/2^{20}$
2	336	3	$2^{20}/2^{20}$	$0/2^{20}$
4	672	5	$2^{20}/2^{20}$	$0/2^{20}$
8	1344	7	$2^{20}/2^{20}$	$0/2^{20}$

Table 2: Reported historical execution counts. Unconditioned totals: filed issue [6]; conditioning cells: execution report [4, Section 2.3], on both scalar and native x86 dispatch. Each observed hit had $Z = 0$. The word-zero control had no hits under either conditioning. These counts are not the proof of the lower bound.

The experiment generated keys using AES counter-mode streams seeded from the operating system. This is execution evidence; the theorem assumes genuinely independent uniform words, not a pseudorandom family. The callable `advanced::V1<3>` through `V4<3>` return the three words and reach the defect. The README/example primarily expose the Style wrappers, which always instantiate width 2.[4]

TODO: Attach the primary unconditioned sampling logs for the 4/3/5/7 counts and the independent harness package. The supplied filed issue records these totals; the supplied execution report does not contain their detailed result rows.

4.2 The valid distance-2 encoder and sharper core bounds

`Encode2` retains six symbols and adds their XOR. One changed symbol changes its parity too; two or more changed symbols already give distance at least two. Thus its distance is exactly 2, on every lane. Its combine matrix is

$$T_2 = \begin{pmatrix} 1 & 0 & 1 & 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 & 1 & 4 & 1 \end{pmatrix}.$$

The general minor method gives singleton sum 5 and full kernel bound 4, but actual NH has a stronger truncated-output property.[9, 10, 5]

Lemma 4.1 (Truncated NH). *For equal-length distinct inputs and every fixed target c , the NH difference modulo either 2^{62} or 2^{63} has probability at most 2ε .*

Proof. Condition on all but a differing pair and use (1). Let $d_x, d_y \in [0, Q]$ be its half-word differences. First use modulus $M = Q^2/4$. If one nonzero difference has valuation at most 30, choose it as d_x and fix its key half. The difference as a function of the other uniform half z has, on each of at most two wrap intervals, slope $c - c'$ with $\gcd(c - c', M) = 2^{\nu_2(d_x)} \leq Q/4$. Solutions within an interval are separated by $M/\gcd(c - c', M) \geq Q$, so there is at most one per interval, or two in total.

The remaining case has $d_x, d_y \in \{0, Q/2\}$, with at least one nonzero. Choose $d_x = Q/2$. The solution period is $Q/2$. If $d_y = 0$, a single interval of length Q contains at most two solutions. If $d_y = Q/2$, the two wrap intervals each have length $Q/2$ and contain at most one each. This proves the bound modulo 2^{62} . Modulo 2^{63} the bound also follows because specifying that residue implies specifying its residue modulo 2^{62} . [5] $\square$

Select two differing encoded symbols. For their leaf output differences D_1, D_2 , including arbitrary fixed targets from conditioned symbols, the actual matrix gives

$$\Pr[D_1 = 0] + \Pr[D_2 = 0] \leq 3\varepsilon, \quad \Pr[D_1 = D_2 = 0] \leq 2\varepsilon^2. \quad (8)$$

Indeed, a row with an odd selected coefficient costs ε , and a row with only even selected coefficients costs at most 2ε by Lemma 4.1; at least one row has an odd coefficient. For the joint bound, determinants of valuation at most one use Lemma 3.1. The only pairs of valuation two are columns $\{0, 5\}$ and $\{1, 6\}$ (zero based), triangular systems with a coefficient 4. The equation with that coefficient costs 2ε modulo 2^{62} ; the other equation fixes an independently keyed NH difference with cost ε . This proves (8), including SIMD by choosing a differing lane. [5, 9]

Set $s_h = (1 - \varepsilon)^{h+1}$ and $a_h = 1 - s_h$. The same conditional product argument as before now proves

$$B_2(h) = (a_h + s_h\varepsilon)(a_h + 2s_h\varepsilon) \leq (h+2)(h+3)2^{-64}. \quad (9)$$

It bounds collisions of the equal-length 16-byte core whenever a group exists. No groups, or different raw tails, give ε^2 . This improves the earlier $(h+2)(h+5)2^{-64}$ bound in the proof note. [5, 9]

For completeness the corresponding defective-core bounds in the fourth summary also follow from the same argument. Every column of T_3, T_4 has an odd coefficient and a second nonzero coefficient of valuation at most two. Choose these two rows for a differing symbol and condition on other symbol keys. Their equality probabilities sum to at most 3ε , while joint equality costs at most ε through the odd coefficient. Hence valid common upper bounds are

$$B_3(h) = B_4(h) = \varepsilon + a_h\varepsilon + (1 - 2\varepsilon)a_h^2 \leq 2^{-32} + (h+1)(h+2)2^{-64}, \quad (10)$$

$$B_5(h) \leq (h+2)2^{-32}. \quad (11)$$

For T_5 use one odd coefficient and one coordinate's forest bound. Here B_k denotes an envelope, not an exact collision probability. Differing tails or no groups cost ε^k . [5]

TODO: Obtain the full fourth-review code-bound proof and certificates, including its truncated-NH derivation and reported 289 encoder rank checks. Equations (9)–(11) are reconstructed above; the unavailable package has not been inspected.

4.3 Actual forest depth and the stack-safe domain

The implementation promotes a full stack level *before* inserting the next leaf. Its live counts are therefore bijective base-eight digits: for $N > 0$, digits lie in $\{1, \dots, 8\}$ with no zero holes. If the highest level is h , then

$$\frac{8^{h+1} - 1}{7} \leq N \leq \frac{8^{h+2} - 8}{7}, \quad h(N) = \lfloor \log_8(7N + 1) \rfloor - 1.$$

This follows either by summing the smallest and largest digit strings, or by induction on insertion. It differs from the paper's ordinary base-eight forest; the coarser $\lfloor \log_8 N \rfloor$ used in the earlier proof note is a safe upper bound. [9, 5]

The stack has nine levels, but the finalizer scans until a zero sentinel without checking the array bound. Consequently the largest contiguous safe domain is

$$N < \frac{8^9 - 1}{7} = 19,173,961, \quad h \leq 7, \quad \text{at most 64 live roots.} \quad (12)$$

At the last safe count the digits are eight eights followed by the zero sentinel; at the next count there are nine ones and the next sentinel read is out of bounds. For output width k , group size is $G_k = 24d_k b$ bytes with $d_2 = 6, d_3 = d_4 = 7, d_5 = 5$. Require $\lfloor n/G_k \rfloor$ to satisfy (12). [9, 10, 5]

Wrapper	b	Exclusive byte limit	Last possible core-key index
Style64	1	2,761,050,384	805
Style128	2	5,522,100,768	952
Style256	4	11,044,201,536	1246
Style512	8	22,088,403,072	1834

Table 3: Stack-safe wrapper domains and absolute key indices, from [9, Section 2], independently checked in [10]. Every wrapper returns 64 bits; its Style number is the block width.

Equations (9)–(11) with $h \leq 7$ give uniform equal-length bounds: raw 16-byte core $< 90 \cdot 2^{-64}$ (about 57.51 bits), raw 24/32-byte cores $< 2^{-32} + 72 \cdot 2^{-64}$, and raw 40-byte core $< 9 \cdot 2^{-32}$. These are restricted-domain pointwise bounds, separate from length-normalised scores.[5]

4.4 The wrapper’s key overlap is compatible with a proof

Number the caller’s entropy words $K[0], K[1], \dots$. Length tables occupy words 0–2047; the two core-output tables occupy 2048–4095 and 4096–6143. The core nevertheless starts at word 512. EHC uses words 512–532; tree keys reserve words 533–658; the finalizer starts at 659, after nine reserved levels regardless of actual depth. With r live roots, root keys occupy $2br$ words; the raw tail uses at most $19b$ distinct words, including its always-inserted padded block. Thus every core read is below

$$659 + b(2 \cdot 64 + 19) = 659 + 147b \leq 1835 < 2048.$$

There is overlap with the length tables, but none with the output tables. The public entropy allocation supplies 8866 words (70,928 bytes).[9, 10]

Let F_x, F_y be the two-word core signatures, $C = \{F_x = F_y\}$, and D the XOR difference of their length-table values. Condition on *all* words below 2048, fixing F_x, F_y, D together. If C fails, a differing core byte exposes a fresh output-table entry, so wrapper equality has probability exactly δ , whatever D is. If C holds, output tables cancel and equality holds exactly when $D = 0$. Therefore

$$\Pr[H(x) = H(y)] = \delta \Pr[\neg C] + \Pr[C \cap \{D = 0\}]. \quad (13)$$

For equal lengths this is $\delta + (1 - \delta) \Pr[C]$. For unequal lengths, ordinary tabulation gives the *marginal* identity $\Pr[D = 0] = \delta$; it need not be independent of C . Since $\Pr[C] \geq \Pr[C \cap \{D = 0\}]$, (13) yields

$$\Pr[H(x) = H(y)] \leq \delta + (1 - \delta) \Pr[D = 0] = 2\delta - \delta^2. \quad (14)$$

Lengths differing in their low 16 bits have probability exactly δ : a differing length byte exposes a fresh entry in rows zero or one, before the core’s starting address. This argument does not assume independent core and length hashes.[9, 10] The uniform stack-safe wrapper bound, now including unequal lengths, is $< 91 \cdot 2^{-64}$ (about 57.49 bits), using (9) and $h \leq 7$. [5]

4.5 Optimal length-normalised coefficient: 63 bits

Theorem 4.2. *Fix one Style wrapper and a positive integer L such that $8L$ bytes are within its domain in Table 3. For any distinct messages of at most $8L$ bytes,*

$$\Pr[H(x) = H(y)] \leq L(2^{-63} - 2^{-128}) = L(2 - 2^{-64})2^{-64}. \quad (15)$$

The coefficient $2 - 2^{-64}$ is optimal across these caps.

Proof. Unequal lengths satisfy (14), which is at most the right side for $L \geq 1$. At equal lengths below a group, the raw-tail bound is δ , so (13) gives $2\delta - \delta^2$ again. At height $h \geq 0$, use the weaker envelope from (9) to get $\delta + (1 - \delta)c_h\delta$, where $c_h = (h + 2)(h + 3)$. A group uses $18b$ words, and this height requires

$$L \geq 18b \frac{8^{h+1} - 1}{7}.$$

At $h = 0$ this is $L \geq 18b$, while $c_0 + 1 = 7$. As h increases, the height threshold grows by more than eight and $c_h + 1$ grows by less than eight. Thus $c_h + 1 \leq L$ throughout. Since $1 + (1 - \delta)c_h \leq c_h + 1 \leq L \leq L(2 - \delta)$, the desired bound follows. These comparisons hold throughout the safe domain.[5, 9]

For optimality take the one-byte messages `0x00` and `0x01`. Only lane zero of one raw pair differs. In each core coordinate its difference vanishes exactly when the relevant high key half is zero, by (7). The relevant absolute key words `659` and `659 + b` are independent, so $\Pr[C] = \varepsilon^2 = \delta$. Length terms cancel, giving the exact wrapper collision probability $\delta + (1 - \delta)\delta = 2^{-63} - 2^{-128}$. This pair has cap $L = 1$ and forces the stated coefficient.[9, 10, 5] $\square$

In the supplied write-up’s metric $\inf_L \log_2(L/\eta(L))$, the optimal family score is $64 - \log_2(2 - 2^{-64})$, slightly above 63; its conservative integer certificate is 63 bits. This is not a constant 2^{-63} collision bound for every length, nor a claim of a 128-bit output.[9, 12, 5]

4.6 Raw lengths and portability

The advanced cores contain no length encoding. Empty input and one zero byte both yield the same zero-padded tail block, no full groups, and identical core outputs for *every* key. Consequently the raw-core bounds above are equal-length theorems. The wrapper length tables resolve this particular pair, with collision probability δ .[9, 5]

The literal `TabulateAfter` declares a three-row table view at width two, yet takes rows eight and sixteen. A sufficiently large allocation does not make those subscripts valid ISO C++; UBSan diagnoses the defect. SIMD horizontal sums also require modular rather than undefined signed-overflow semantics. The existing tests use disclosed compiler options, including `-fwrapv`, where needed.[9, 4]

Stock NEON dispatch refers to undeclared `V2Neon`, `V3Neon`, `V4Neon`, although shared NEON/SSE2 wrappers are named `V2Sse2`, etc. The supplied fix uses the shared SSE2-or-NEON guard and removes the obsolete NEON branch. Earlier NEON tests used explicit dispatch-name aliases; they do not show that the stock header compiled. The supplied PR reports agreement for 4609 input vectors over 14 builds, with 73,744 output words per build; this establishes tested equivalence, not absence of undefined behaviour.[7, 8]

5 What remains open and suggested fixes

The intended theorem requires the distance assumption actually to hold. Restore the encoders’ row traversal and capture semantics, with the loops also restricted to initialized data symbols; merely changing the capture is insufficient. A concrete distance-3 alternative is systematic x_i plus $P = \bigoplus_i x_i$ and $Q' = \bigoplus_i A_i x_i$ over $\mathbb{F}_2^3$, using the matrices with column encodings

$$(1, 2, 4), (4, 5, 2), (5, 7, 6), (2, 6, 5), (3, 4, 1), (6, 3, 7), (7, 1, 3).$$

Each A_i and $A_i + A_j$ is invertible (the audit supplies 28 rank checks). One changed data symbol changes both parities; with two changed data symbols either P changes, or their differences coincide and $(A_i + A_j)$ changes Q' . Three changed data symbols already give distance three. This is a proved replacement design, not a declaration that the shipped routine is fixed.[2]

Correct the table representation, signed sums, NEON dispatch and stack scan; document supported lengths and the actual key-access budget. The advanced sizing helper underestimates some short width-3 calls because the finalizer offset reserves fixed tree levels; fix it against the access map. Preserve equal-length scope for raw cores or add an injective length representation. The present overlap need not be removed to obtain Theorem 4.2, but disjoint domains would simplify future maintenance.[9, 4, 8]

Open work includes checked source-to-arithmetic correspondence for supported compilers, complete machine formalisation, and analysis of any particular seeded entropy expansion. The exact worst-case collision probabilities of the defective long-message cores are not established here. None of these gaps invalidates the restricted arithmetic theorems or the explicit equal-length lower bound. The paper’s original pointwise formula remains unjustified even where its headline can be recovered.[12, 5]

TODO: Obtain the second review’s full paper-errata and replacement-bound proof. Its condensed summary is available; its full third-party proof text is not.

6 Machine-checked status

Lean formalisation: *in progress*. The requested `LEAN_HALFTIME_STATUS.md` file was not present at the final status check for this draft. No theorem above is described as Lean-checked on that basis.

The supplied certificates and fresh checks are executable evidence, not Lean proofs. For this draft, `checks/rederive.py` checks all 441 maximal minors by two exact determinant algorithms, all projection-subset coefficient bounds for these matrices, small-word truncated-NH instances, forest-depth identities, the 28 corrected-encoder ranks and the wrapper coefficient comparisons. A symbolic real-header check confirms the parity dependencies of `Encode3/4/5`. A rebuilt scalar harness includes the unchanged supplied header and rechecks the exact EHC witness, full-output propagation on favourable fibres, and wrapper address formula. Historical scalar/SSE2/AVX-512 and disclosed NEON-alias logs remain separately identified. The original harness has 7680 selected key/context cases per width, including 1280 favourable cases; these are deterministic execution checks, not sampled collision estimates.[9, 10]

A Paper errata and resource accounting

This list combines the audit, second review and filed issue; their formulae are re-derived here.[2, 11, 6, 5]

1. Definition 1 must require $x \neq y$. For identical inputs, difference zero has probability one.
2. The page-five pigeonhole argument must include the zero column when counting binary column types. Nonsingularity is a property of each selected square submatrix over the rationals; it does not imply invertibility modulo 2^{64} .
3. Theorem 1 needs $t = \max \nu_2(\det T_F)$, the missing division, and an upper-bound inequality, as in Section 2.
4. The page-eight expression $m\varepsilon^k$ should be $(m\varepsilon)^k$ in the independent-coordinate tree calculation. This correction does not repair Lemma 3's missing coordinate-difference condition.
5. Equation (3)'s binary forest needs $\lfloor \lg n \rfloor + 1$ slots; the recurrence and displayed codomain have inconsistent lengths.
6. Section 2 counts n in 32-bit words, but Section 6.1 counts b in 64-bit words. The group count is $N = \lfloor n/(2bdw) \rfloor$ in the former units.
7. For the intended forest, root count is $r_f(N) \leq (f-1)(h+1)$, not fh . With $N = 63, f = 8, h = 1$, there are 14 roots, exceeding 8. The $b f h k$ final-NH seed budget therefore fails for some forests.
8. A full f -ary forest has $(N - r_f(N))/(f-1)$ internal nodes. Each needs $f-1$ modified-NH products per coordinate/lane, so tree multiplications total $kb(N - r_f(N))$. Adding EHC and root-finalizer work gives $b(ew + k)N$ products. If the tail contains ℓ 64-bit pairs, the total is

$$M = b(ew + k)N + k\ell, \quad I = ew + k(f-1)h + kb r_f(N) + (\ell + k - 1)\mathbf{1}_{\ell > 0}$$

seed words for the intended scalar-pair-shift Toeplitz allocation. Here the b factor in EHC work counts physical lanes. This is not the code's fixed nine-level reservation, block-shift tail layout or always-inserted padding rule.

9. The paper's own seed-size expression, with its HalftimeHash24 parameters, evaluates to $197 + 213h$ words. At $h = 3$ and $h = 16$ this is 6688 and 28,840 bytes, rather than the stated 8.4 KB and 34 KB. Correcting the forest budget also changes that resource formula; these arithmetic substitutions alone are not a repaired allocation theorem.
10. The leading EHC multiplication fraction is $ew/(ew + k)$: $27/30 = 90\%$ for HalftimeHash24, whereas $27/32$ gives the rounded 84% for HalftimeHash40. This is an operation count, not a new performance measurement.
11. The introduction's $O(n^{2-c})$ expected-collision assertion needs $\varepsilon = O(n^{-c})$ when applied through the pair-count bound.

B Displayed matrices and numerical provenance

Besides T_2 in Section 4, the paper’s matrices are [1, 11]

$$T_3 = \begin{pmatrix} 0 & 0 & 1 & 4 & 1 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 0 & 1 & 4 & 1 & 2 & 2 \\ 1 & 4 & 1 & 1 & 0 & 0 & 2 & 1 & 2 \end{pmatrix},$$

$$T_4 = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 4 & 2 & 4 & 1 & 1 \\ 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 4 & 1 \\ 2 & 0 & 1 & 0 & 4 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 4 & 1 & 2 & 8 \end{pmatrix}, \quad T_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 0 & 0 & 1 & 2 & 1 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 & 3 & 8 & 5 \\ 0 & 0 & 0 & 1 & 0 & 1 & 4 & 9 & 8 \\ 0 & 0 & 0 & 0 & 1 & 1 & 5 & 3 & 9 \end{pmatrix}.$$

Maximal-valuation examples (zero-based columns) are $(0, 5)$ with determinant 4 for T_2 ; $(0, 1, 3)$ with determinant 12 for T_3 ; $(0, 1, 5, 6)$ with determinant -24 for T_4 ; and $(0, 1, 2, 4, 8)$ with determinant -8 for T_5 . [11]

All substantive numerical claims above trace to the supplied files: paper parameters and matrices to the saved paper; determinant values and intended bounds to the second and fourth summaries; indices, byte limits, fibre counts and the sharp short pair to `PROOF.md` and the wrapper certificates; sampling totals to the filed issue; conditioned counts to the execution report; portability-build counts to the NEON PR. The fresh derivation log is `checks/rederived.json`. This log checks the arithmetic rather than creating an unsupported source for a historical measurement.

C Evidence qualifications

The audited header’s SHA-256 is

7ef5dd48f54537b430f85bc1867b23a93551ab1c56415cfcef362d1651956cc3.

The saved paper is arXiv:2104.08865v2. Inputs are under `../inputs/`; source copies and certificates are in `halftime-review-support/`. The full second/fourth review packages, the sampling harness `verify2/`, and the NEON PR’s larger vector corpus are referenced by their summaries but are not all supplied here. [9, 4, 5, 8]

Two source inconsistencies matter. First, the referee’s suggested “first six blocks” correction and the filed issue’s shorthand “symbol 0 XOR symbol 1” are superseded by the execution report’s exact block masks, reproduced in (6); the parity packets use blocks zero through three. Second, the execution report’s concluding “iff” and its inconsistent conditioned 2^{34} wording are not adopted: its detailed conditioning table says 2^{20} trials per cell, while the proof note establishes a sufficient full-hash event, not necessity. Table 2 follows the detailed cells and the filed issue’s unconditioned totals. The theorem uses exact fibre counting, so it is independent of those report inconsistencies. [9, 10, 4, 6]

TODO: Reconcile the execution report’s final-section wording with its raw logs, and archive the NEON PR’s vector corpus and verification package. The present note uses the detailed conditioning table and does not claim to have rerun the large historical experiments.

References

- [1] Jim Apple. HalftimeHash: Modern hashing without 64-bit multipliers or finite fields. arXiv:2104.08865v2, 2021. Saved PDF and text in the supplied source archive.
- [2] Independent second reading of the hash-proof audit, 2026. Supplied analysis record, 18 September; `AUDIT_REVIEW.md, Section3`.
- [3] John Black, Shai Halevi, Hugo Krawczyk, Ted Krovetz, and Phillip Rogaway. UMAC: Fast and secure message authentication. In *Advances in Cryptology – CRYPTO ’99*, volume 1666 of *Lecture Notes in Computer Science*, pages 216–233. Springer, 1999.
- [4] Independent execution-based verification of two HalftimeHash claims, 2026. Supplied analysis record, 18 September; `halftime_execution.md`.

- [5] Fourth HalftimeHash review: repaired paper theorem and code-level bounds: condensed, 2026. Supplied analysis record, 18 September; `fourth_review_summary.md`.
- [6] Filed HalftimeHash proof and implementation issue text, 2026. Supplied analysis record, 18 September; `halftime-issue-final.md`.
- [7] arm64 NEON dispatch references undeclared wrappers, 2026. Supplied analysis record, 18 September; `ISSUE.md`.
- [8] Fix arm64 NEON dispatch to use the existing shared wrappers, 2026. Supplied analysis record, 18 September; `PR.md`.
- [9] HalftimeHash wrapper and Encode3 certificates, 2026. Supplied analysis record, 18 September; `PROOF.md;halftime-review-support/wrapper-cert/`.
- [10] Adversarial referee report: the HalftimeHash 64-bit wrapper certificate, 2026. Supplied analysis record, 18 September; `halftime_referee.md`.
- [11] Second independent review of the HalftimeHash paper: condensed, 2026. Supplied analysis record, 18 September; `second_review_summary.md`.
- [12] Verdict and summary on the supplied HalftimeHash header, 2026. Supplied analysis record, 18 September; `VERDICT.mdandSUMMARY.md`.